

The Energy Cost of Privacy in Federated Learning

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Abstract—Differential privacy in federated learning is evaluated along two axes, privacy and utility. On battery-powered edge devices a third decides whether a configuration is deployable at all, and it is not reported. We measure the energy cost of example-level differential privacy applied locally at each client, with NVML hardware counters differenced around each federated round, sweeping the privacy budget on UCI-HAR, whose clients are the individual study participants, and on CIFAR-10. The cost separates in two. A fixed per-round overhead from per-sample gradients and clipping costs +51% on HAR and +34% on CIFAR-10, with no dependence on ϵ that a permutation test over seeds resolves ($p = 0.27$ and $p = 0.79$); everything budget-dependent acts through the number of rounds required, and is the larger term. Measured as energy-to-target-accuracy, privacy is free at 0.40 accuracy on HAR for $\epsilon \geq 1$, $5.7\times$ at 0.70, and unbounded at 0.85, which no budget we tested reaches at any expenditure. Because the cost is paid in joules it inherits the local carbon intensity: the premium attributable to privacy alone is 106–148 gCO₂eq per 1000 runs on Algeria’s grid against 5.7–7.9 g in Norway, a quantity the privacy-utility framing cannot express. We also report a measurement hazard: running the privacy accountant’s noise calibration inside the timed loop reproduced an entire apparent ϵ trend.

Keywords—federated learning, differential privacy, energy efficiency, green AI, carbon footprint, edge computing, human activity recognition

I. INTRODUCTION

Federated learning allows many devices to train a shared model without sending their raw data anywhere. That property is often described as privacy preserving, but on its own it is not. Model updates leak information about the data that produced them, and gradients alone are enough to recover training examples. Differential privacy closes this gap with a formal guarantee, and it has become the standard defense [1], [2].

The guarantee is not free. Its cost is measured in accuracy, and the privacy-utility trade-off is now well mapped: a budget ϵ is chosen, the resulting accuracy is reported, and the trade-off between them is the result [1], [3]. What is not reported is what the choice costs to run.

This matters because federated learning exists to serve battery-powered devices. A privacy configuration a phone cannot afford to execute is not a worse point on the privacy-utility curve; it is not on the curve at all. Green AI established energy and carbon as quantities worth measuring [4], [5], but

attention has gone to large centralized training, where one operator owns the hardware and can move the job to a cleaner grid. Federated learning inverts both conditions.

Previous work approaches this and stops short. Wireless resource-allocation studies minimize device energy under a privacy constraint [6]–[10], but the energy there is an objective inside a convergence limit rather than something measured and ϵ is fixed by the problem statement rather than varied. Convergence theory under client-level privacy proves that an optimal number of aggregations exists and that training past it degrades the model [11]–[13]. Those results say where to stop; they do not say what continuing costs, because a round is an abstract unit in them. Close to us, Schwermer et al. [14] instrumented a physical edge testbed and reported energy with differential privacy enabled against disabled in a single privacy setting.

What none of them produces is the shape of the cost across the budget. A designer choosing ϵ needs to know whether halving it doubles the energy bill or puts the accuracy target completely out of reach, and those are different decisions.

We measure that shape. Energy comes from NVML [15] (NVIDIA Management Library) hardware counters for each federated round, not from device power ratings, and the privacy budget is accounted over the whole run rather than per round. The guarantee is example-level, applied locally at each client so that it holds against an untrusted server, with the budget held per client over the whole run. Our primary dataset is UCI-HAR [16], where the federation’s clients are the individual volunteers who contributed the data, so the client boundary is a real person rather than a synthetic partition. CIFAR-10 checks the same effects on a harder benchmark.

The paper makes four claims. The cost of privacy separates into a fixed per-round overhead from per-sample gradients and clipping and a convergence penalty that acts through the round count; the two are different sizes and call for different remedies. Measured against the accuracy a deployment actually needs, the cost is not a multiplier but a ceiling above which no expenditure buys the target. Under a tight budget, training past a certain round spends energy and loses accuracy at the same time, so the stopping point is set by the privacy budget, rather than by the task. Because the cost is paid in joules, the premium attributable to privacy is a geographic quantity,

which the privacy-utility framing cannot express.

One further result concerns the measurement itself. Deriving the noise multiplier requires a search over the privacy accountant, and that search costs more for a loose budget than a tight one. Running inside the federated loop, where the natural implementation places it, its cost arrives inside the measurement window and scales with ϵ , taking the shape of a privacy cost. In one of our sweeps it produced the entire observed trend, and nothing in the resulting numbers announced the problem. Any energy study of a privacy mechanism is exposed to this, so we describe it and the controls that remove it.

II. RELATED WORK

A. Measuring the energy of training

Green AI made energy and carbon reportable quantities rather than externalities [4], [5], and a family of tools now estimates them from utilization traces and grid factors [17]–[19]. Comparisons of those tools against instrumented hardware find the estimates diverge substantially from measurement [20], [21], which is why we read NVML counters rather than model power from device ratings. Patterson et al. [22] established that where a job runs changes its emissions by 5–10 $\times$; we apply that to the premium a privacy budget adds.

B. Energy and differential privacy in federated learning

One line of work treats DP noise as a constraint inside a wireless resource-allocation problem and minimizes device energy under it, either by harvesting channel noise as the privacy mechanism [6]–[8] or by co-designing compression with the noise budget [9], [10]. Energy there is an optimization objective evaluated through a convergence bound; it is not measured and ϵ is a constraint rather than a swept variable. Closest to us empirically, Schwermer et al. [14] built a physically distributed edge testbed and reported traffic, energy and training time for a CNN on MNIST with differential privacy enabled, and found software estimates deviating by up to 35% from measurement. Their contrast is DP on versus off at one setting; ours is the shape of the cost across the budget. Aroussi et al. [23] schedule ϵ adaptively on Raspberry Pi and ESP32 nodes and report a 20–25% energy improvement over a fixed-privacy baseline, the gain of a controller, rather than the cost curve. A designer would need to choose the budget in the first place.

C. Rounds as the mechanism

That the cost should act through the round count is known analytically. Wei et al. [11] prove that under client-level DP there is an optimal number of aggregations for a given protection level, beyond which convergence degrades, and later derive a discounting method to find it [12]; the same optimum reappears for personalized FL [24], for query and reply counts [13], and in control formulations that adapt noise and rounds jointly [25]. These results say where to stop. What they do not say is what continuing costs, because the round is an abstract unit in them. Our measurements price it: on CIFAR-10 at $\epsilon = 0.5$ the peak this literature predicts falls

at round 8 of 30, and the rounds after it are priced in joules below.

D. Our position

We therefore contribute the missing empirical axis: hardware-counter energy swept across ϵ under whole-run per-client accounting [1], decomposed into a per-round overhead and a convergence penalty, and reported as energy-to-target-accuracy rather than energy per round or energy per run. Our guarantee is example-level and local, weaker than the client-level protection of [2], [11]; the energy question is orthogonal to that choice, and the per-round overhead we measure is the cost of the per-sample gradients that both require.

III. METHODOLOGY

A. Federated Setup

Both datasets run on Flower [26] with FedAvg, 50% of clients sampled per round for training and 50% for evaluation, one local epoch, batch 32, SGD at lr 0.01 with momentum 0.9, three seeds per configuration, and $\epsilon \in \{0.5, 1, 2, 4, 8, \infty\}$ at $\delta = 10^{-5}$. Neither model uses batch normalization, which Opacus [3] rejects because it mixes information across examples.

UCI-HAR [16] is primary. We keep the published subject-disjoint split: its 21 training participants become the federation’s clients, one per person, and the 9 test participants are held out, so central accuracy measures generalization to people the federation never saw. The partition is given by who contributed the data rather than by a heterogeneity parameter. We train on the raw inertial signals (nine channels, 128 samples per window) with a three-block 1D CNN and global average pooling (37k parameters) for 100 rounds. Local train and validation sets are split within each activity class, in file order, discarding the straddling window: windows overlap by 50%, so a random split would leak, and the activities come in blocks, so one ordered cut would drop classes.

CIFAR-10 is secondary: a five-layer CNN (62k parameters) across 20 clients over 30 rounds, partitioned by a Dirichlet distribution over labels at $\alpha = 0.5$ [27].

The HAR round budget comes from the non-private convergence curve rather than from convenience. Under whole-run accounting the number of rounds enters the privacy calculation directly, so it is not a free parameter.

B. Privacy Accounting

We enforce *example-level* differential privacy, applied locally: Opacus [3] clips and noises per-sample gradients inside local training, so the guarantee protects one of a client’s own examples, the noise is added before anything leaves the device, so it holds against an untrusted server, and each client holds its own (ϵ, δ) budget over the whole run. This is weaker than the client-level guarantee of [2], [11], which noises the entire client update and conceals participation itself; the per-round overhead we measure is the per-sample gradient work both need. The budget is accounted over every local SGD step across every round a client joins, not per round and not per

epoch. For a client with n examples, batch B , R rounds, sampling fraction q_c and E local epochs:

$$s_{\text{total}} = R q_c E \lceil n/B \rceil \quad (1)$$

We invert the RDP accountant [28] once, before training, to obtain the noise multiplier σ giving the target ε after s_{total} steps, and hold it constant. Precomputing is deliberate: in simulation each client is reconstructed every round, so an engine-internal accountant resets each round and reports a per-round ε while the true budget compounds. Opacus samples at a rate $1/\lceil n/B \rceil$ rather than B/n , and we calibrate at the rate the sampler uses. We claim no amplification by client subsampling, which would require secure aggregation or a shuffler.

The q_c in (1) makes s_{total} an *expected* participation count, and nothing in the protocol bounds the realised one, which is binomial. So σ is calibrated optimistically: recomputing each client’s spent ε from the participation log, the worst client overspends in every private run we measured, by 4–22% (Table I, last column). A guarantee that must hold unconditionally has to drop q_c and calibrate against $R E \lceil n/B \rceil$, or cap participation per client; we report the target ε as the design parameter it is, with the realised worst case beside it.

Because per-client dataset sizes differ, σ differs per client, and the two datasets sit in different regimes: a HAR client holds far fewer examples, which raises the sampling rate, and runs a longer schedule. Both push σ up.

C. Energy Measurement

Energy comes from the NVML counter, which accumulates millijoules on the device. Differencing it around each round requires no power sampling and no integration and cannot miss a transient. Measurement is server-side; all clients share one GPU in simulation, so a round-level window captures the full cost regardless of actor scheduling, whereas measuring inside each client would double-count concurrent work.

Every energy figure in this paper is net of the idle baseline unless a column is labelled Gross. We report both, the baseline sampled per run after board power settles, since a card does not reach its floor the moment a run ends and probing during that tail reads a busy baseline. Baselines are checked against the sweep median, and outliers are discarded. Net energy subtracts the baseline over the wall clock the counter covers, excluding start-up, central evaluation and teardown. The three seeds also give measurement noise floors, each the worst within-condition spread across seeds and each measured on the quantity it bounds — energy per round, board power, final accuracy — and reported in Section IV; no smaller difference is treated as an effect. Where a claim is about an ordering across budgets rather than a single difference, a floor is the wrong instrument, because the quantity on trial is the spread of the condition means: there we permute the seed labels across the five private budgets and report the p -value of the resulting F .

D. Experimental Controls

We fix the clipping norm $C = 1.0$ for all ε values to isolate the effect of privacy. We verify each round that the

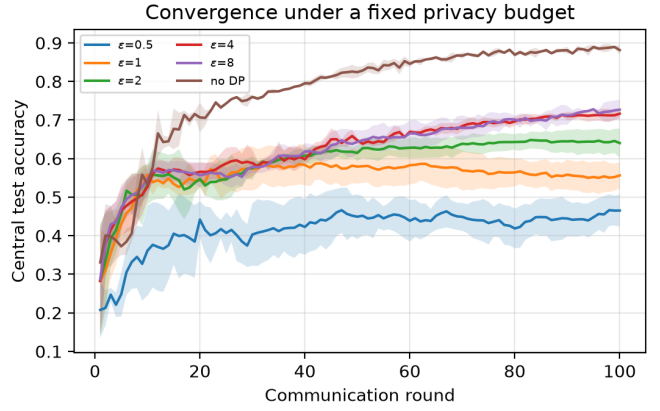


Fig. 1. Central test accuracy per round on UCI-HAR, mean ± 1 SD over three seeds. No stopping round is marked: on HAR no peak-to-final drop clears the across-seed spread.

number of optimizer steps matches the privacy accountant’s calibration. All noise scales σ are computed before timing, since the accountant’s search time varies with ε and could otherwise create a false energy trend. We randomize the order of ε values to separate privacy effects from execution-order drift. We do not force deterministic GPU kernels, as this can change kernel selection and energy use. Instead, we assess reproducibility across multiple random seeds.

E. Limitations

Clients are virtual and share one GPU, although on HAR each holds one real participant’s data. CPU RAPL counters were unavailable, so measurement is GPU-only and communication energy is modeled. The workload does not saturate the device, which is why net figures accompany gross ones. Per-channel standardization constants come from the training split and are global across clients, so a real deployment would have to agree them without pooling raw data. And the HAR round budget is set from the non-private convergence curve, so private conditions inherit a longer schedule than any of them needs and a larger σ with it; their accuracies are a lower bound on what a per- ε tuned budget would give.

IV. RESULTS

All figures are over three seeds and net of the idle baseline. Noise floors are the worst within-condition across-seed spread of the quantity each one bounds: energy per round, 12.0% on HAR; board power, 7.8% on HAR and 8.9% on CIFAR-10; final accuracy, ± 0.051 and ± 0.031 . No smaller difference is resolvable.

A. The Per-Round Cost Does Not Track the Budget

On HAR, private training consumes 21.1 J per round against 14.0 J without privacy, an overhead of +51% (Table I, Fig. 2). The overhead combines +26.7% wall-clock and +19.0% mean power, so a power ratio alone would report 19%. CIFAR-10 gives +34%.

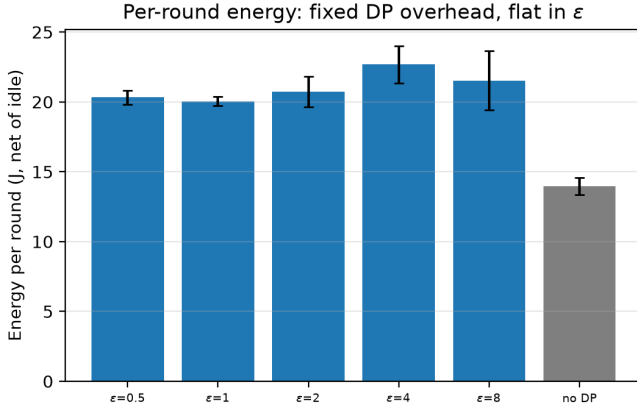


Fig. 2. Net energy per round on UCI-HAR, mean ± 1 SD over three seeds; non-private baseline in grey.

TABLE I
PER-ROUND ENERGY ON UCI-HAR, MEAN OF THREE SEEDS. OVERHEAD IS NET J/ROUND AGAINST $\epsilon = \infty$; WORST ϵ IS WHAT THE MOST-SAMPLED CLIENT ACTUALLY SPENT.

ϵ	Gross W	Net W	Net J/rd	Overhead	Worst ϵ	σ range
0.5	59.66	42.13	20.3	+45.4%	0.570	17.50–20.62
1	59.27	41.60	20.0	+43.4%	1.122	9.22–10.94
2	61.13	43.01	20.7	+48.3%	2.285	4.94–5.90
4	65.77	47.18	22.7	+62.2%	4.250	2.75–3.25
8	62.68	44.71	21.5	+54.0%	9.112	1.64–1.90
∞	54.47	36.75	14.0	—	—	—

Whether that overhead depends on ϵ is a question about five condition means, not one difference, so a noise floor is the wrong instrument. The per-condition overheads run from +43.4% to +62.2%, a spread of 12.5% of their mean, and the across-seed spread *within* one budget is the same size, ± 2.58 J/round or 12.0%: at $\epsilon = 4$, the largest overhead, the three seeds span 20.9–24.1 J. A permutation test on the fifteen per-seed values gives $F = 1.46$, $p = 0.27$; on CIFAR-10, $F = 0.46$, $p = 0.79$. We report no resolvable dependence on ϵ and claim no equality: three seeds bound any budget effect at about the size of the overhead’s own scatter, not at zero.

These results separate the two sources of energy cost: the per-round overhead comes from the DP operations and does not order with ϵ , while everything budget-dependent acts through the number of rounds a target accuracy needs, which per-round energy cannot see.

B. The Cost Grows with the Target

Energy per round suggests that privacy adds a fixed cost, but the total cost rises sharply with the target accuracy (Table II). Every entry in Table II is a mean over all three seeds except the one marked ‡, which two of three reached. At 0.40 accuracy, privacy adds no measurable cost for $\epsilon \geq 1$ (187–210 J against 185 J without it). The private runs reach this first target in fewer rounds than the non-private one, which we attribute to noise acting as regularization early in training; the effect reverses at every higher target.

TABLE II
NET ENERGY (J) TO A TARGET ACCURACY ON UCI-HAR; MEAN ROUNDS IN PARENTHESES. “—”: UNREACHED IN 100 ROUNDS. ‡: TWO OF THREE SEEDS. INCLUDES A ≈ 110 J WARM-UP ROUND.

ϵ	Target accuracy			
	0.40	0.50	0.70	0.85
0.5	357 (13.0)	334‡ (12.0)	—	—
1	201 (4.7)	258 (7.7)	—	—
2	197 (4.0)	353 (12.0)	—	—
4	210 (4.3)	253 (6.3)	1710 (74.3)	—
8	187 (3.3)	269 (7.7)	1699 (76.3)	—
∞	185 (6.0)	197 (7.0)	301 (15.0)	839 (57.0)

At 0.50, private training requires $1.28\text{--}1.79\times$ the non-private energy, rising to $5.7\times$ at 0.70; $\epsilon \leq 2$ does not reach this target. At 0.85, the non-private model reaches the target in 57.0 rounds and 839 J, while no tested private budget reaches it.

C. An Energy-Optimal Stopping Round

Under a sufficiently tight budget the accuracy curve peaks and declines, so training past the peak spends energy and loses accuracy at once. We report a stopping round only where the peak-to-final drop exceeds that condition’s across-seed spread and the seeds agree on where the peak falls.

On CIFAR-10 it resolves at the two tightest budgets. At $\epsilon = 0.5$ the mean peaks at 0.176 at round 8 and ends at 0.150, a drop of 0.026 against ± 0.010 , the seeds peaking at rounds 8, 8 and 9; the rounds after the peak account for 60–68% of the run’s net energy. At $\epsilon = 1$ the drop is 0.044 against ± 0.014 , with 42–61% of the energy spent after a peak the seeds place between rounds 10 and 16. At $\epsilon \geq 2$ the drops fall inside their spreads. Those energy fractions are close to the fraction of the schedule that follows the peak, because per-round energy is flat; the result is the peak’s position and the resolvable decline after it, while the joules price the tail of a round budget the designer chose. The actionable form is the peak round itself: round 8 of 30 at $\epsilon = 0.5$.

On HAR (Fig. 1) no budget resolves one over 100 rounds. Seeds do peak early and give accuracy back—at $\epsilon = 1$ one peaks at round 10 and loses 0.079, spending 85% of its energy afterwards—but the peak rounds disagree (10, 33 and 39 at $\epsilon = 1$; 16, 54 and 65 at $\epsilon = 0.5$) and the drop in the mean stays inside the ± 0.042 spread of final accuracy. The mechanism is visible per run; its location does not survive averaging.

The mechanism is the same on both—accumulated noise eventually dominates the aggregated update—but where the crossover falls depends on the task and, on HAR, on the seed. So we state it conditionally: a tight enough budget induces an energy-optimal stopping round well before the budget expires, and training for a fixed round count regardless of ϵ risks burning energy while degrading the model. We claim no ordering of the peak round in ϵ .

D. CIFAR-10 as a Secondary Check

CIFAR-10 reproduces the per-round result (+34% net of idle, spread 6.0% inside an 8.9% floor) and carries the

stopping effect. Reaching 0.20 accuracy costs 1076 J net at $\varepsilon = 1$ against 677 J without privacy, $1.59\times$, because the private run needs 8.0 rounds against 3.7; at $\varepsilon = 0.5$ only two of three seeds reach it. Absolute accuracies are low (0.490 without privacy, 0.15–0.30 under it), as expected for a 62k CNN over 30 non-IID rounds, which is why HAR carries the headline claims.

E. Ablations

We vary one factor at a time at $\varepsilon \in \{1, \infty\}$ with three seeds per cell. On HAR only local epochs and the clipping norm vary: the federation size is fixed by the data split and there is no Dirichlet parameter. The ablation sweep ran in its own session, whose baseline sits about 2 J/round above Table I’s, so energies here are compared within this sweep only.

Under a tight budget, the model becomes noise-limited.

At $\varepsilon = 1$ on HAR, neither local epochs nor clipping norm moves final accuracy outside the ± 0.041 across-seed floor of these cells. With $\sigma \approx 10$, injected noise dominating these settings, tuning them does not improve the private model.

The usual efficiency lever stops paying. Without privacy, increasing local epochs from one to five reduces the energy needed to reach 0.70 accuracy from 345 J to 193 J, a $1.79\times$ saving. At $\varepsilon = 1$, however, the target is not reached at any epoch count, while per-round energy rises from 22 J to 49 J. More local steps increase s_{total} , which raises σ and offsets the potential convergence benefit.

On CIFAR-10, the effect reverses. At $\varepsilon = 1$, accuracy drops from 0.182 to 0.114 to 0.100 as local work increases, well beyond the ± 0.024 accuracy floor of these cells. The peak round falls from 15 to 6, while post-peak energy waste rises from 1785 J to 7312 J.

Clipping affects convergence, not computation. Changing C leaves per-round energy essentially unchanged on both datasets, but it can affect convergence, particularly on CIFAR-10. For example, at $C = 2$ it peaks at 0.187 but falls to 0.100 by the end, losing 46.7% of its peak accuracy. Thus, the fixed $C = 1.0$ used in our main experiments is conservative rather than optimal.

F. Carbon Cost by Grid

Table III scales the measured HAR energy to 1000 runs and apply published grid carbon intensities [29]; the scaling repeats the measured workload rather than extrapolating to other hardware. The cost attributable to privacy ranges from 106–148 gCO₂eq on Algeria’s gas-dominated grid to 5.7–7.9 g in Norway. The factor of 18.8 between them is the ratio of the two grid intensities and would be identical for non-private training; what privacy contributes is the premium itself, which is a quantity in grams and therefore geographic.

V. DISCUSSION

A. Privacy Sets a Quality Ceiling

The two costs admit different remedies — the per-round overhead is an implementation problem, the convergence penalty a scheduling one — and per-round energy sees only

TABLE III
CARBON PER 1000 HAR RUNS (GCO₂EQ) AT INTENSITIES OF 26, 56, 344, 369, 488 AND 631 G/KWH [29]. PRIVACY: DP MINUS NO-DP.

Grid	DP	No DP	Privacy
Norway	21	15	6–8
France	46	32	12–17
Germany	284	198	75–104
USA	305	212	80–112
Algeria	403	281	106–148
India	521	363	137–191

the first. For a designer the consequence is that privacy has no single energy multiplier: it is nearly free at a low accuracy target and unreachable at a high one, so the budget and the target have to be chosen together rather than in sequence. The reason is that the round budget is itself spent out of the privacy budget: more participation requires more noise for the same ε , so training longer can stop helping, and a stopping rule is needed even though where it falls depends on the dataset and the budget.

B. When the Model Is Noise-Limited, Tuning Is Not the Lever

Our HAR ablations are negative and useful: a practitioner responding to a disappointing private model by sweeping local epochs or the clipping norm will spend considerable energy to land where they started. This does not hold in every regime, and CIFAR-10 shows why: there the private model retains more signal, and the same settings can be got badly wrong. Increasing local work per round is the standard way to reduce communication in federated learning; under DP it stops paying on HAR and actively harms on CIFAR-10, because local steps enter the budget and force a larger σ .

C. Where the Computation Runs

The geographic point is sharper for federated learning than for centralized training. The computation sits on whatever grids the participating devices do, and no operator chooses them. A deployment cannot relocate the way a data center can; what it can adjust is how much computation the privacy requirement demands.

D. Measuring Energy Around a Privacy Accountant

Instrumentation attached to a privacy mechanism tends to scale with the privacy parameter, so its cost mimics the effect under study. The accountant’s noise-calibration search is the case we hit: running inside the federated loop, it reproduced our entire observed ε trend, while executed step counts stayed flat, and nothing in the resulting numbers announces this. The three controls of Section III-D remove it, and we would apply them to any energy study of a privacy mechanism.

E. Threats to Validity

Clients are virtual and share one GPU, so the ratios transfer to real devices, but the absolute joules do not; confirming them on physical hardware is the immediate next step. Measurement is GPU-only and communication energy is modelled — for a 0.15–0.24 MB model over tens of rounds we expect it small,

but have not shown it. Two datasets bound the effects rather than establishing them. Our guarantee is example-level and local: a client-level mechanism would noise the whole update instead, changing the accuracy cost but not the per-sample gradient work that produces the per-round overhead. And we claim no amplification by subsampling, so these are costs under weak trust assumptions, not the best achievable — while in the other direction the realised participation count puts the worst client above its nominal budget in every private run.

VI. CONCLUSION

We measured the energy cost of differential privacy in federated learning using hardware counters around each round on UCI-HAR and CIFAR-10.

The cost has two components. A fixed per-round overhead from per-sample gradients and clipping adds 51% on HAR and 34% on CIFAR-10, in neither case ordered in ε . The larger cost comes from the additional rounds required to reach a target accuracy: reaching 0.70 on HAR requires $5.7\times$ the non-private energy at $\varepsilon = 4$, while $\varepsilon \leq 2$ does not reach the target.

Three practical implications follow. First, set the round budget from the privacy budget rather than the task, since additional rounds consume privacy budget and can become counterproductive. At $\varepsilon = 0.5$ on CIFAR-10 the peak falls at round 8 of 30, and the rounds after it carry two thirds of the run's energy. Second, choose the privacy budget and target accuracy jointly because privacy can impose an accuracy ceiling. Third, when the model is noise-limited, hyperparameter tuning provides little benefit.

Finally, energy measurements can be biased by privacy-accounting overhead. We therefore recommend precomputing noise multipliers, verifying step counts, and randomizing run order. Future work should identify when the system shifts from implementation-limited to noise-limited behavior.

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